

A02

Q1: Statistics Proof

$$\begin{aligned} 1) \quad m(a + bX) &= a + b(m(X)) \\ m(a) + m(bX) &= a + b(m(X)) \\ a + b(m(X)) &= a + b(m(X)) \quad \checkmark \end{aligned}$$

$$\begin{aligned} 2) \quad \text{Cov}(X, a + bY) &= b \times \text{Cov}(X, Y) \\ \text{Cov}(X, a) + \text{Cov}(X, bY) &= b \times \text{Cov}(X, Y) \\ 0 + \text{Cov}(X, bY) &= b \times \text{Cov}(X, Y) \\ b \times \text{Cov}(X, Y) &= b \times \text{Cov}(X, Y) \quad \checkmark \end{aligned}$$

$$\begin{aligned} 3) \quad \text{Cov}(a + bX, a + bX) &= b^2 \text{Cov}(X, X) \\ \text{Cov}(a + bX, a) + \text{Cov}(a + bX, bX) &= b^2 \text{Cov}(X, X) \\ \text{Cov}(a, a) + \text{Cov}(a, bX) + \text{Cov}(a, bX) + \text{Cov}(bX, bX) &= b^2 \text{Cov}(X, X) \\ 0 + 0 + 0 + \text{Cov}(bX, bX) &= b^2 \text{Cov}(X, X) \\ b \cdot b \text{Cov}(X, X) &= b^2 \text{Cov}(X, X) \\ b^2 \text{Cov}(X, X) &= b^2 \text{Cov}(X, X) \quad \checkmark \end{aligned}$$

Pt 2: $\text{Cov}(X, X) = s^2$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (x_i - m(X)) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

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$$4) m(g(x)) = g(m(x))$$

No it is not always.

$$P(X=0) = \frac{1}{2}, P(X=2) = \frac{1}{2}$$

$$E[X] = (0 \cdot \frac{1}{2}) + (2 \cdot \frac{1}{2}) = 1$$

$$\text{Let } g(x) = x^2 \quad g(E[X]) = g(1) = 1^2 = \boxed{1}$$

$$E(g(X))$$

$$X=0 \rightarrow g(0) = 0^2 = 0$$

$$X=2 \rightarrow g(2) = 2^2 = 4$$

$$E(g(X)) = (0 \cdot \frac{1}{2}) + (4 \cdot \frac{1}{2}) = \boxed{2}$$

$$2 \neq 1$$